

# ISI-BUDS Lab Day 14

July 10th 2026

Unadjusted Model:  $E[Y] = \beta_0 + \beta_1 X$

Adjusted Model:  $E[Y] = \gamma_0 + \gamma_1 X + \gamma_2 W$

Proposition 1:

$$E[\hat{\beta}_1] = \gamma_1 + r_{XW} \sqrt{\frac{\text{Var}(W)}{\text{Var}(X)}} \gamma_2$$

Proposition 2:

$$\text{Var}(\hat{\gamma}_1) = \frac{\sigma_{Y|X}^2 - \gamma_2^2 \text{Var}(W|X)}{n \text{Var}(X)(1 - r_{WX}^2)}$$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma_{Y|X}^2}{n \text{Var}(X)}$$

In this lab you will conduct a simulation study to confirm Propositions 1 & 2 in order to investigate the impact of omitting confounders or important predictors of response on inference regarding some predictor of interest in linear regression. Let  $X \sim N(0, \sigma_X^2)$  be the predictor of interest,  $W \sim N(0, \sigma_W^2)$  with  $\text{corr}(X, W) = \rho$  be the confounder or the nonconfounding important predictor. Note that  $W$  will tend to be a confounder when  $\rho \neq 0$  and  $\gamma_2 \neq 0$ , and it will tend to be a nonconfounding predictor of interest (precision variable) when  $\rho = 0$  and  $\gamma_2 \neq 0$ . Let  $Y_i \sim N(\mu_i, \sigma^2)$  be the response such that

$$\mu_i = \gamma_0 + \gamma_1 X_i + \gamma_2 W_i, \quad i = 1, \dots, n.$$

For each of the cases listed below, we will simulate datasets (at least 1000) and perform linear regressions of  $Y$  on  $X$  alone and of  $Y$  on  $X$  and  $W$ . For each case given below, compare the results from the two regression models with respect to (1) the probability of rejecting  $H_0 : \gamma_1 = 0$ , (2) the bias of  $\hat{\gamma}_1$ , and (3) the standard errors of  $\hat{\gamma}_1$ .

Write a simulation study to complete the tables below for the case when we regress  $Y$  on  $X$  alone and when we regress  $Y$  on  $X$  and  $W$ . For your simulation, take  $\sigma_X^2 = 1.25$  and  $\sigma_W^2 = 2$ . Note that the package `mvtnorm` may be useful if you are writing your simulation in R. Comment on your results in terms of how they agree with Propositions 1 & 2 and the relevance to choosing adjustment variables in practice.

Table 1: Unadjusted Model Results

| Scenario | $\gamma_1$     | $\gamma_2$     | $\rho = \text{corr}(X, W)$ | $\Pr[\text{Reject } H_0 : \gamma_1 = 0]$ | Bias of $\hat{\gamma}_1$ | $\text{SE}[\hat{\gamma}_1]$ |
|----------|----------------|----------------|----------------------------|------------------------------------------|--------------------------|-----------------------------|
| 1        | $\gamma_1 = 0$ | $\gamma_2 = 0$ | $\rho = 0$                 |                                          |                          |                             |
| 2        | $\gamma_1 = 0$ | $\gamma_2 = 0$ | $\rho = 0.7$               |                                          |                          |                             |
| 3        | $\gamma_1 = 0$ | $\gamma_2 = 2$ | $\rho = 0$                 |                                          |                          |                             |
| 4        | $\gamma_1 = 0$ | $\gamma_2 = 2$ | $\rho = 0.7$               |                                          |                          |                             |
| 5        | $\gamma_1 = 1$ | $\gamma_2 = 0$ | $\rho = 0$                 |                                          |                          |                             |
| 6        | $\gamma_1 = 1$ | $\gamma_2 = 0$ | $\rho = 0.7$               |                                          |                          |                             |
| 7        | $\gamma_1 = 1$ | $\gamma_2 = 2$ | $\rho = 0$                 |                                          |                          |                             |
| 8        | $\gamma_1 = 1$ | $\gamma_2 = 2$ | $\rho = 0.7$               |                                          |                          |                             |

Table 2: Adjusted Model Results

| Scenario | $\gamma_1$     | $\gamma_2$     | $\rho = \text{corr}(X, W)$ | $\Pr[\text{Reject } H_0 : \gamma_1 = 0]$ | Bias of $\hat{\gamma}_1$ | $\text{SE}[\hat{\gamma}_1]$ |
|----------|----------------|----------------|----------------------------|------------------------------------------|--------------------------|-----------------------------|
| 1        | $\gamma_1 = 0$ | $\gamma_2 = 0$ | $\rho = 0$                 |                                          |                          |                             |
| 2        | $\gamma_1 = 0$ | $\gamma_2 = 0$ | $\rho = 0.7$               |                                          |                          |                             |
| 3        | $\gamma_1 = 0$ | $\gamma_2 = 2$ | $\rho = 0$                 |                                          |                          |                             |
| 4        | $\gamma_1 = 0$ | $\gamma_2 = 2$ | $\rho = 0.7$               |                                          |                          |                             |
| 5        | $\gamma_1 = 1$ | $\gamma_2 = 0$ | $\rho = 0$                 |                                          |                          |                             |
| 6        | $\gamma_1 = 1$ | $\gamma_2 = 0$ | $\rho = 0.7$               |                                          |                          |                             |
| 7        | $\gamma_1 = 1$ | $\gamma_2 = 2$ | $\rho = 0$                 |                                          |                          |                             |
| 8        | $\gamma_1 = 1$ | $\gamma_2 = 2$ | $\rho = 0.7$               |                                          |                          |                             |

Comment below on your results in terms of how they agree with Propositions 1 & 2 and the relevance to choosing adjustment variables in practice.